

human judgment, critical thinking, and persuasion. This interrelation will result in a lower AI exposure of journalism routine tasks than those of technical writers.

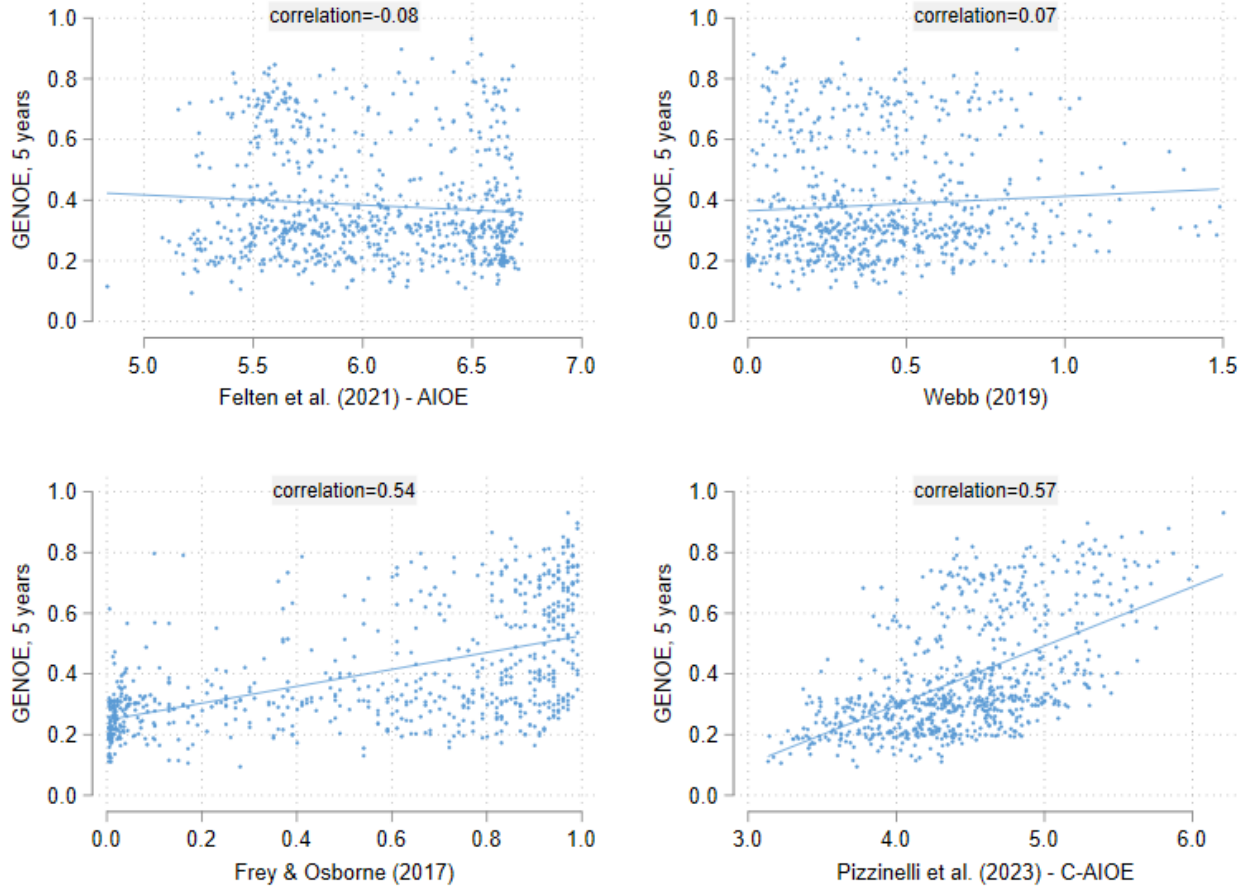
Another limitation of this approach is that it does not account for the ethical and social context of different occupations performing similar types of tasks. Consider, for example, judges and credit analysts. Both make decisions based on evaluating extensive information, and both roles can significantly impact individuals' lives. However, the nature and scope of their impact differ in important ways. Judges deliver verdicts that can dramatically affect individuals' well-being, freedom, rights, and long-term societal standing. Their role demands a profound understanding of ethical principles and the ability to balance complex cases. The ethical weight and social impact of their decisions, in addition to requiring human judgment and empathy, make it unlikely that societies would accept these roles being legitimately replaced by AI. Credit analysts, while also impacting individual well-being through their decisions on loan approvals or credit ratings, typically operate within a more narrowly defined financial context. Their decisions can certainly affect a person's economic opportunities and quality of life, but generally do not carry the same weight of personal liberty or broader societal impact as judicial decisions. This comparison highlights the importance of considering not just the technical similarities in task types, but also the broader ethical, social, and institutional contexts in which different occupations operate when assessing the potential for AI integration or replacement.

Our approach corrects the limitations of previous indexes by integrating a more holistic task-based analysis that captures the interrelations and complementarities between tasks within occupations. Unlike previous methods that disaggregate occupations into isolated tasks, skills, or abilities, our AI-generated index considers the entire vector of tasks, reflecting a complete view of occupational contributions in production. To address the second limitation, we provide the model in the surveys with both the name of the occupation and its vector of tasks, allowing it to estimate the overall likelihood of AI replacement without imposing restrictive assumptions. This method allows the model to consider the ethical and social context of each occupation organically rather than through predefined criteria, offering a more holistic evaluation of exposure and replacement. Alternative prompt specifications and follow-up surveys suggest that our method effectively accounts for these considerations.

This second concern is also raised by Pizzinelli *et al.* (2023) and is somewhat addressed by their complementarity-adjusted occupational exposure index (C-AIOE). This measure incorporates factors such as work contexts and job zones to account for the likelihood that AI will complement rather than replace human labor.⁴ Our approach goes further by imposing fewer assumptions

⁴Notwithstanding its advancements over previous indexes, the work contexts and job zones retrieved from the O*NET contain several limitations and assumptions. Firstly, it assumes that higher levels of education and training automatically correlate with better integration of AI, which may not account for the specific skill sets and practical contexts required in each occupation. Additionally, the chosen contexts, such as face-to-face communication, responsibility for others' health, and physical proximity, might not fully capture the ethical reasoning required in certain roles, nor the degree of public resistance to AI replacing certain decision-making roles. Furthermore, as the index is built upon the AIOE, it relies solely on the isolated similarity of abilities to AI developments and thus does not account for the exposure of tasks and their interrelations in the workplace.

Figure 2: Correlation of GENOE and Other Indexes of Occupational Exposure



Source: Authors' formulation based on data from O*NET Felten *et al.* (2021); Webb (2019); Pizzinelli *et al.* (2023). Note: This figure displays the GENOE for the next one, five, and 10 years. The index measures the likelihood that occupations will be displaced by AI within each time horizon.

3.3 Alternative Methodologies

We explore how GENOE estimates vary under different methodologies and prompt choices. Our baseline specification involves providing the model with the occupation name and a vector of tasks and asking for an estimate of the likelihood of job replacement. We then introduce six alternative indexes, each employing a distinct approach:

Blind Index: This approach removes the occupation name, relying solely on the task vector for estimations. This allows us to isolate the impact of task-specific factors from occupation-specific context factors in the model's predictions and analyze if the absence of the occupation name significantly alters the estimates.

Task-by-Task Index: Instead of aggregating tasks under an occupation, we estimate the likeli-

hood of replacement for each task individually. These estimates are then averaged and weighted according to the relevance and importance of each task within the occupation, providing a granular view of task vulnerability.

Ethical Emphasis Index: Here, we explicitly ask the model to consider the ethical repercussions and potential social resistance to replacing certain tasks with AI. This approach highlights whether explicitly including ethical considerations causes a significant deviation from the baseline, which may already implicitly factor in such considerations.

Ethical Exclusion Index: This index explicitly asks the model not to consider ethical, regulatory, or social impediments to replacing tasks with AI. The objective of this index, compared to the previous one, is to measure the increase in the index when these considerations are excluded.

LLM-Specific Index: This variant focuses specifically on large language model (LLM) technologies. By narrowing the scope to LLM innovations, we assess the unique impact of advancements in natural language processing and related applications on occupational exposure, contrasting it with the general AI exposure measured in the baseline.

Non-AI Technologies Index: We estimate the probability of task replacement by non-AI technologies such as digital tools, robotics, and computer software. This provides a comparative baseline, distinguishing the impact of traditional technologies from AI-driven automation.

AI-Only Index: This index is derived by subtracting the Non-AI Technologies Index from the baseline GENOE, capturing the incremental exposure attributable solely to AI advancements. It allows us to identify occupations where AI-specific risks have significantly increased.

Table 4 provides the descriptive statistics of these alternative indexes, their correlation with the baseline GENOE, and other existing exposure indexes, and the extent to which they differ from the baseline. Both the Blind Index and the Ethical Emphasis Index do not deviate significantly from the baseline, although the mean differences are statistically significant. This indicates that there are slight changes in the reasoning of the model when the occupation name is omitted or when ethical considerations are explicitly prompted, but most of these factors are already incorporated in the baseline index. The overall correlations of these indexes with the baseline GENOE are 0.95 and 0.99, respectively. This consistency is supported by follow-up surveys, where ethical considerations and occupation-specific criteria were crucial elements for many of the model's assessments.

GENOE's ability to reflect ethical, regulatory, and social considerations in its assessment of AI's potential to replace human labor, even when not explicitly directed to do so, is a critical aspect of our argument. This aspect was initially confirmed by our follow-up survey analysis and is also supported by the comparisons with alternative indexes. Notably, when we explicitly instructed the model to include these factors, the resulting index showed only a minor reduction of one percentage point over the 5-year horizon. However, when we asked the model to exclude these considerations, the index increased by four percentage points, from 0.38 to 0.42. A similar pattern is observed for the 10-year horizon, as illustrated in Appendix Figure A.3, though the effect is less